

Indices and surds - Set 2 - Solutions

Specification MM1.1, MM1.2, M2.7, M2.11, M2.12.

Answer key

Question	Answer	Topic
1	D	Indices and surds
2	F	Indices and surds
3	E	Indices and surds
4	F	Indices and surds
5	D	Indices and surds
6	B	Indices and surds
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8	F	Indices and surds
9	A	Indices and surds
10	H	Indices and surds
11	D	Indices and surds

Worked solutions

Question 1 IND-20 · Indices and surds · spec MM1.2

Given that $\sqrt{a} - \sqrt{b} = 2$ and $a - b = 12$, find $\sqrt{a} + \sqrt{b}$.

Answer: D

$$a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b}), \text{ so } 12 = 2(\sqrt{a} + \sqrt{b}), \text{ giving } 6.$$

Question 2 IND-21 · Indices and surds · spec MM1.1

If $x = 2^p$, $y = 2^q$ and $\frac{x^3}{y^2} = 2^5$ with $xy = 2^{10}$, find p .

Answer: F

$$3p - 2q = 5 \text{ and } p + q = 10. \text{ From the second, } q = 10 - p; \text{ then } 3p - 20 + 2p = 5, \text{ so } 5p = 25 \text{ and } p = 5.$$

Question 3 IND-06 · Indices and surds · spec MM1.2

Given that $(2 + \sqrt{3})^2 = a + b\sqrt{3}$ and $(2 + \sqrt{3})^{-1} = c + d\sqrt{3}$, where a, b, c, d are integers, find the value of $a + b + c + d$.

Answer: E

$$(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}, \text{ so } a = 7, b = 4. (2 + \sqrt{3})^{-1} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}, \text{ so } c = 2, d = -1. \text{ Sum} \\ = 7 + 4 + 2 - 1 = 12.$$

Question 4 IND-07 · Indices and surds · spec MM1.1

Which of the following is equal to $\frac{x^2\sqrt{x}}{\sqrt[3]{x}}$ for $x > 0$?

Answer: F

$$\text{The index is } 2 + \frac{1}{2} - \frac{1}{3} = \frac{12+3-2}{6} = \frac{13}{6}.$$

Question 5 IND-09 · Indices and surds · spec MM1.2

Find the value of $\frac{1}{\sqrt{2} + 1} + \frac{1}{\sqrt{3} + \sqrt{2}} + \frac{1}{2 + \sqrt{3}}$.

Answer: D

$$\text{Rationalising each term gives } (\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (2 - \sqrt{3}), \text{ which telescopes to } 1.$$

Question 6 IND-11 · Indices and surds · spec MM1.2

Given that $x = \sqrt{7} + \sqrt{3}$ and $y = \sqrt{7} - \sqrt{3}$, find $x^2 + y^2 - xy$.

Answer: B

$$x^2 + y^2 = (10 + 2\sqrt{21}) + (10 - 2\sqrt{21}) = 20 \text{ and } xy = 7 - 3 = 4. \text{ So } 20 - 4 = 16.$$

Question 7 IND-16 · Indices and surds · spec MM1.2

Which of the following statements is/are true for all positive real numbers a and b ?

- I $\sqrt{a}\sqrt{b} = \sqrt{ab}$
 II $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$
 III $\sqrt{a^2 + b^2} < a + b$

Answer: F

I is a law of surds. II fails for $a = b = 1$: $\sqrt{2} \neq 2$. III: $(a+b)^2 = a^2 + b^2 + 2ab > a^2 + b^2$ since $ab > 0$, and both sides are positive, so III is true.

Question 8 IND-18 · Indices and surds · spec MM1.2

The expression $\frac{3}{7-2\sqrt{3}} - \frac{3}{7+2\sqrt{3}}$ simplifies to

Answer: F

Common denominator $(7-2\sqrt{3})(7+2\sqrt{3}) = 49 - 12 = 37$. Numerator: $3(7+2\sqrt{3}) - 3(7-2\sqrt{3}) = 12\sqrt{3}$.

Question 9 IND-22 · Indices and surds · spec MM1.2

Which one of the following is the largest?

Answer: A

Each option is $\sqrt{a} + \sqrt{b}$ with $a + b = 9$. Squaring gives $9 + 2\sqrt{ab}$, which is largest when the product ab is largest, i.e. when a and b are as close as possible. Option E has $a = b = 4.5$ (product 20.25), beating $4 \times 5 = 20$ for option C. Numerically: E $= \sqrt{18} \approx 4.243$, C $= 2 + \sqrt{5} \approx 4.236$.

Question 10 X2-12 · Indices and surds · spec MM1.1

a and b are positive real numbers. Which of the following statements is/are true?

- I If $a^2 = b^2$ then $a = b$.
 II If $2^a = 2^b$ then $a = b$.
 III If $a^b = 1$ then $a = 1$.

Answer: H

I holds because $a, b > 0$ rules out $a = -b$. II holds because 2^x is one-to-one. III: for positive a and b , $a^b = 1$ gives $b \log a = 0$, so $\log a = 0$ and $a = 1$. All three are true.

Question 11 IND-12 · Indices and surds · spec MM1.1

For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?

Answer: D

For the sum to be rational, both $\sqrt{n}$ and $\sqrt{n+1}$ would need to be rational (if one is irrational, $\sqrt{n} + \sqrt{n+1}$ squared gives $2n + 1 + 2\sqrt{n(n+1)}$ and $n(n+1)$ is never a perfect square for $n \geq 1$ since $n^2 < n(n+1) < (n+1)^2$). No two consecutive positive integers are both squares, so the answer is 0.